Python Activity-Based Learning
Program 10: Campus Event Financial Planning

Problem Statement

Calculate campus event expenses for venue, equipment, catering, transport, and certificates. Determine participant contributions, sponsorship shortfall, and the minimum registration count needed to break even.

Objectives

• Calculate the total event budget.

• Calculate sponsorship shortfall and participant contribution.

• Find the minimum number of registrations needed to break even.

Assumptions

• All listed costs are fixed event costs except catering, which is charged per participant.

• The registration fee per participant is fixed.

• Sponsorship is a fixed amount; minimum registrations are calculated using the remaining amount divided by the contribution per participant.

Algorithm

1. Read venue, equipment, transport, and certificate costs.

2. Read catering cost per participant, sponsorship, and registration fee.

3. Calculate fixed costs and total cost for a given expected participant count.

4. Calculate sponsorship shortfall.

5. Calculate participant contribution and minimum break-even registrations.

6. Display the financial summary.

Flowchart

```
START → Read event costs → Read participant and sponsorship details → Calculate fixed costs → Calculate total
```

Pseudocode

```
BEGIN
  READ venue, equipment, transport, certificate_cost
  READ catering_per_person, expected_people, sponsorship, fee
  fixed_cost = venue + equipment + transport
  total_cost = fixed_cost + certificate_cost + catering_per_person * expected_people
  shortfall = maximum of 0 and (total_cost - sponsorship)
  contribution = fee
  IF fee > catering_per_person
    minimum_people = ceiling(shortfall / (fee - catering_per_person))
  ELSE
    minimum_people = impossible
  DISPLAY total cost, shortfall, minimum people
END
```

Python Source Code

```
venue = float(input("Venue cost: "))
equipment = float(input("Equipment cost: "))
transport = float(input("Transport cost: "))
certificate_cost = float(input("Total certificate cost: "))
catering_per_person = float(input("Catering cost per participant: "))
expected_people = int(input("Expected number of participants: "))
sponsorship = float(input("Sponsorship amount: "))
registration_fee = float(input("Registration fee per participant: "))

if (venue < 0 or equipment < 0 or transport < 0
        or certificate_cost < 0 or catering_per_person < 0
```

```
        or expected_people < 0 or sponsorship < 0
        or registration_fee < 0):
    print("Invalid input.")
else:
    fixed_cost = venue + equipment + transport + certificate_cost
    total_cost = fixed_cost + catering_per_person * expected_people
    shortfall = max(0, total_cost - sponsorship)

    print("\nCAMPUS EVENT FINANCIAL REPORT")
    print("Fixed costs: Rs.", round(fixed_cost, 2))
    print("Total event cost: Rs.", round(total_cost, 2))
    print("Sponsorship: Rs.", round(sponsorship, 2))
    print("Sponsorship shortfall: Rs.", round(shortfall, 2))

    contribution_per_person = registration_fee - catering_per_person

    if shortfall == 0:
        minimum_people = 0
        print("Sponsorship covers the event cost.")
    elif contribution_per_person <= 0:
        minimum_people = -1
        print("Break-even is not possible with this registration fee.")
    else:
        minimum_people = int(shortfall / contribution_per_person)
        if shortfall % contribution_per_person != 0:
            minimum_people = minimum_people + 1
        print("Minimum registrations needed to cover shortfall:",
              minimum_people)

    expected_income = sponsorship + registration_fee * expected_people
    balance = expected_income - total_cost
    print("Expected final balance: Rs.", round(balance, 2))
```

Test Case

Venue ■10,000; equipment ■5,000; transport ■2,000; certificates ■1,000; catering ■100/person; expected participants 100; sponsorship ■10,000; registration fee ■300.
Expected: fixed costs ■18,000; total cost ■28,000; shortfall ■18,000; contribution after catering ■200/person; minimum registrations 90.

Sample Output

```
CAMPUS EVENT FINANCIAL REPORT
Fixed costs: Rs. 18000.0
Total event cost: Rs. 28000.0
Sponsorship: Rs. 10000.0
Sponsorship shortfall: Rs. 18000.0
Minimum registrations needed to cover shortfall: 90
Expected final balance: Rs. 12000.0
```

Conclusion

The program estimates campus event expenses, calculates the sponsorship shortfall, and determines the minimum registrations required to cover the remaining cost.